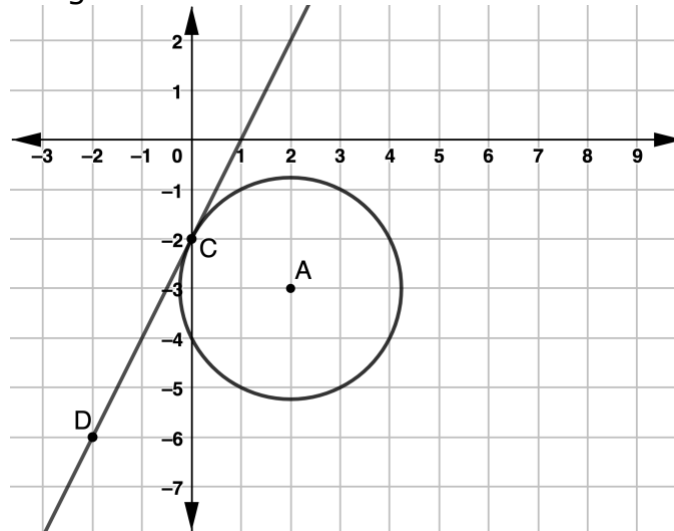


Geometry
Unit 13
Lesson 2 Practice

Name _____
Date _____
Period _____

Use the following information for questions 1-4. Circle A is shown, where point C is on the circle and line CD is tangent to circle A .



1. Use a straightedge to draw a line that goes through points A and C .
2. Write the equation of line CD .
3. Write the equation of line AC .
4. Line CD and line AC intersect at point $(0, -2)$. Use the equations of line CD and AC to confirm this point.

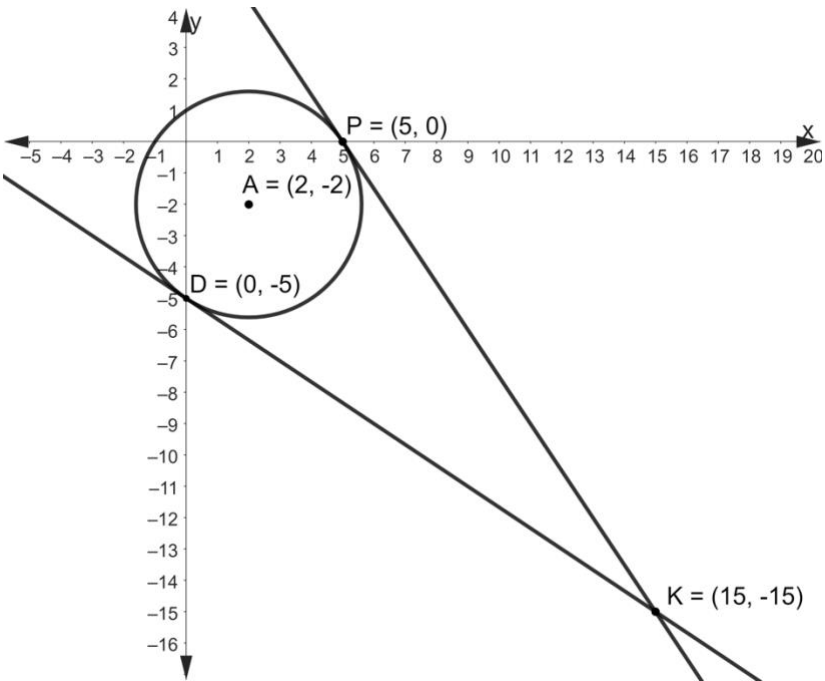
Use the following information for questions 5-8. Circle P has a center at $(4, -3)$ and a tangent line with the equation $y = 4x - 2$.

- 5. Write the equation of the line perpendicular to the tangent line.
- 6. Find the point where the tangent line touches circle P . Call the point H .
- 7. What is the radius of circle P ?
- 8. Write the equation of circle P .

Use the following information for questions 9-10. Circle A is shown with tangent lines PK and DK .

- 9. Complete the table.

Line Segment	Length
$\overline{PK}$	
$\overline{DK}$	
$\overline{AD}$	



- 10. Write the equation of circle A .